$$\frac{\Gamma \vdash e : \tau \to \mathsf{bool}}{\Gamma \vdash \mathsf{flat}(e) : \mathsf{con}(\tau)}\ \text{Type-C-Flat} \qquad \frac{\Gamma \vdash \kappa_1 : \mathsf{con}(\tau_1) \quad \Gamma \vdash \kappa_2 : \mathsf{con}(\tau_2)}{\Gamma \vdash \kappa_1 \to \kappa_2 : \mathsf{con}(\tau_1 \to \tau_2)}\ \text{Type-C-Fun}$$

$$\frac{\Gamma \vdash \kappa_1 : \mathsf{con}(\tau_1) \quad \Gamma \vdash \lambda x : \tau_1 . \kappa_2 : \tau_1 \to \mathsf{con}(\tau_2)}{\Gamma \vdash \kappa_1 \to_i \lambda x . \kappa_2 : \mathsf{con}(\tau_1 \to \tau_2)}\ \text{Type-C-DepFun}$$

$$\frac{\Gamma \vdash \kappa : \mathsf{con}(\tau) \quad \Gamma \vdash e : \tau}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa, e) : \tau}\ \text{Type-C-Mon} \qquad \frac{}{\Gamma \vdash \mathsf{blame}_l^k : \tau}\ \text{Type-C-Blame}$$

$$\frac{\Gamma \vdash e : \mathsf{bool} \quad \Gamma \vdash v : \tau}{\Gamma \vdash \mathsf{check}_l^k(e, v) : \tau}\ \text{Type-C-Check} \qquad \frac{\Gamma \vdash v : \tau \quad \Gamma \vdash \kappa : \mathsf{con}(\tau)}{\Gamma \vdash \mathsf{guard}_j^{k,l}(\kappa, v) : \tau}\ \text{Type-C-Guard}$$

Figure 1: Type Rules for Contracts